Stress-Testing Product-Gated Clustering and Bounded Priority Ranking for Flood-Rescue Reports: A Synthetic Study

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ISDS 2026 | Paper 6444 | Synthetic stress-testing study

Presentation overview

From noisy reports to measured scheduling consequences

01 **Problem and research gap**
Why consolidation errors matter

02 **Data and proposed pipeline**
Two separate synthetic experimental suites

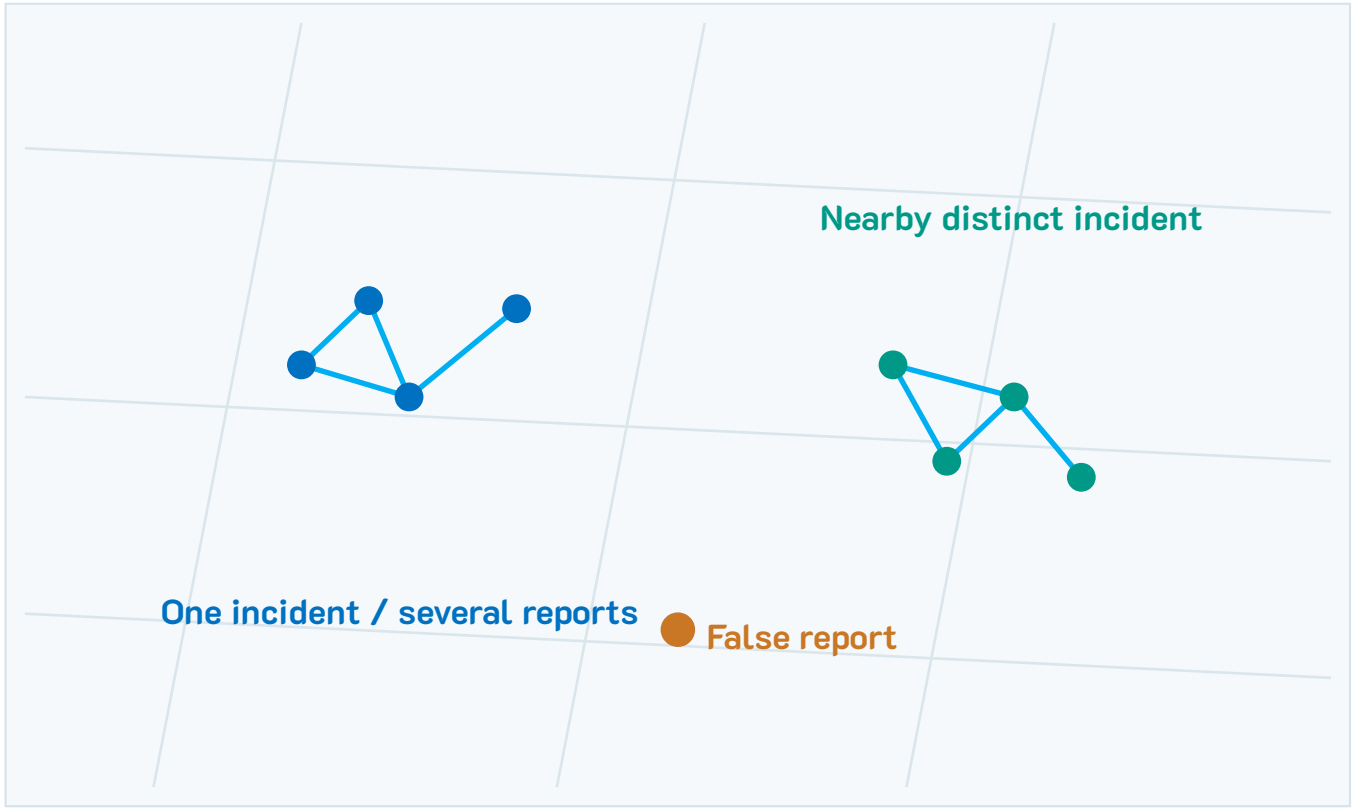
03 **Methods and evaluation**
Product / Additive, Louvain, priority, reported metrics

04 **Results for RQ1–RQ3**
Clustering → ranking → dispatch

05 **Conclusions and next validation**
What the evidence supports and what remains open

Many reports do not mean many incidents

Repeated, nearby and false reports create different operational risks



Repeated evidence

One event can generate many overlapping reports.

False merges / splits

Nearby events may be merged; one event may be split.

False destinations

Coordinated reports may create a destination with no genuine incident.

Related approaches and the study gap

Three research strands meet at the report-to-dispatch boundary

Social sensing

Event detection and crisis-information extraction

[6–8]

Report consolidation

Graph community detection and density-based clustering

[1]; Paper 6444 §1

Humanitarian dispatch

Competing need, efficiency and equity objectives

[5]

GAP EXAMINED IN THIS STUDY

How do predicted splits, merges and false destinations propagate through ranking into a scheduler that cannot consult incident truth?

Observable pipeline and contribution boundaries

Predictions are fixed before evaluator-only truth is joined

Reports

Location / time
Flood / urgency
Claims / support

→

Sparse graph

Observable affinity
Threshold + top-k

→

Clusters

Louvain / Leiden
Predicted groups

→

Priority

Duplicate families
Bounded score

→

Schedule

Predicted destinations
Resource constraints

↓

Missing location / time
→ manual review

EVALUATOR ONLY

Incident identity · independent benefit · deadline · harm

Generator 3.0: what is anchored, what is simulated?

RQ1 · 80 runs · 16 incidents per run · 26,288 reports

Copernicus EMSR848

Historical flood context
Central Viet Nam

OpenStreetMap

Geographic locations
and anchor features

WorldPop 2025

Population reference
around geographic anchors

CONTROLLED SIMULATION

Latent incidents

16 per run

→

Observable reports

Position / time / F / E / N / V

→

Copies + campaigns

Controlled duplication / attacks

Separate exports: observable algorithm input | evaluator-only incident and report labels

Candidate 4.1: a separate fully synthetic suite

RQ2 / RQ3 · 80 datasets · 30,229 reports · synthetic regional geometry

1 Generate incidents

Geometry, start times, needs and vulnerability are simulated.

2 Generate reports

Noisy observable evidence, exact / near copies and attacks.

3 Reserve outcome truth

Benefit, deadlines, service and harm remain evaluator-only.

Why simulation?

Known linkage and controlled perturbations.

What it cannot establish

Field representativeness or policy validity.

Product and Additive: two edge-weight constructions

A node is a report; an edge weight expresses observable affinity

PRODUCT

$$w_{ij}^{\times} = a_i a_j G_{ij}(\beta T_{ij} + \gamma C_{ij})$$

ADDITIVE

$$w_{ij}^{+} = a_i a_j(\alpha G_{ij} + \beta T_{ij} + \gamma C_{ij})$$

G Spatial proximity

T Temporal proximity

C Flood / urgency similarity

a Location-and-time admission

alpha, beta, gamma:
nonnegative weights

Far apart, similar time/context: Product attenuates the full time–context contribution.

Louvain: turn a weighted graph into communities

The graph construction and the community detector are distinct choices



1 Local moves

Move a node when modularity increases.

2 Aggregate

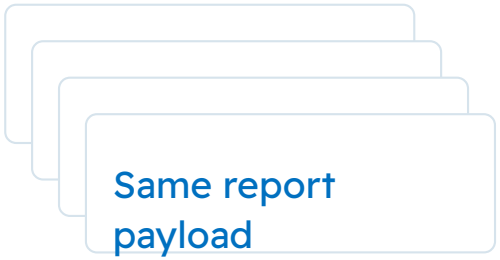
Replace each community with a super-node.

3 Repeat

Continue until no further gain is obtained.

Duplicate-aware, bounded priority

Repeated transport copies should not count as independent evidence



Evidence families

Exact fingerprints and near-duplicate grouping.

Bounded score

Confidence gating, maxima, clipping and saturation.

$$0 \le P_k \le \mu, \qquad 1 \le \mu \le 2$$

Exact-copy invariance requires fixed evidence grouping and score inputs.

Research questions and stage-specific contributions

Each question tests a different part of the pipeline

RQ1	Clustering	Does the independently tuned Product pipeline improve clustering over Additive?	Conditional edge bound + composition / stress comparison.
RQ2	Ranking	Does duplicate-aware priority align with independent simulated benefit under report perturbations?	Exact-copy invariance + alignment and attack audit.
RQ3	Dispatch	How do predicted split, merge and noise errors affect dispatch outcomes?	Paired scheduling audit using predicted destinations.

Evaluation protocol and information boundaries

Select on calibration; hold configurations fixed for the test audit

Generator 3.0 / RQ1

20 development	20 calibration	40 held-out test

Candidate 4.1 / RQ2–RQ3

20 development	20 calibration	40 held-out test

RQ2: oracle incident grouping

Isolate ranking from upstream clustering errors.

RQ3: predicted clusters

Average resource conditions within each seed: 40 pairs.

Reported metrics: clustering

Compare predicted groups with evaluator-only incident linkage

ARI

HIGHER IS BETTER

Adjusted Rand Index: agreement between two partitions, corrected for chance.

Pairwise F1

HIGHER IS BETTER

Treat each pair of reports as same-incident or different-incident; combine precision and recall.

1

Identical partitions

≈ 0

Chance-level agreement

< 0

Possible discordance

Reported metrics: ranking and score stability

Normalized Discounted Cumulative Gain at five (NDCG@5) and score stability

NDCG@5 ↑

$$\mathrm{DCG@5} = \sum_{r=1}^{5} \frac{b_{\pi_r}}{\log_2(r+1)}$$

$$\mathrm{NDCG@5} = \frac{\mathrm{DCG@5}}{\text{Ideal DCG@5}}$$

b: independent simulated benefit

π: order induced by the priority score

1: ideal ranking; larger is better

Priority drift ↓

Absolute normalized score change after a perturbation.

Top-five churn ↓

Change in membership of the five highest-ranked incidents.

Reported metrics: dispatch outcomes

Consequences are evaluated only after the schedule is fixed

Simulated harm

LOWER IS BETTER

Mean incident harm from the synthetic lateness / service model, not measured real-world injury.

Deadline misses

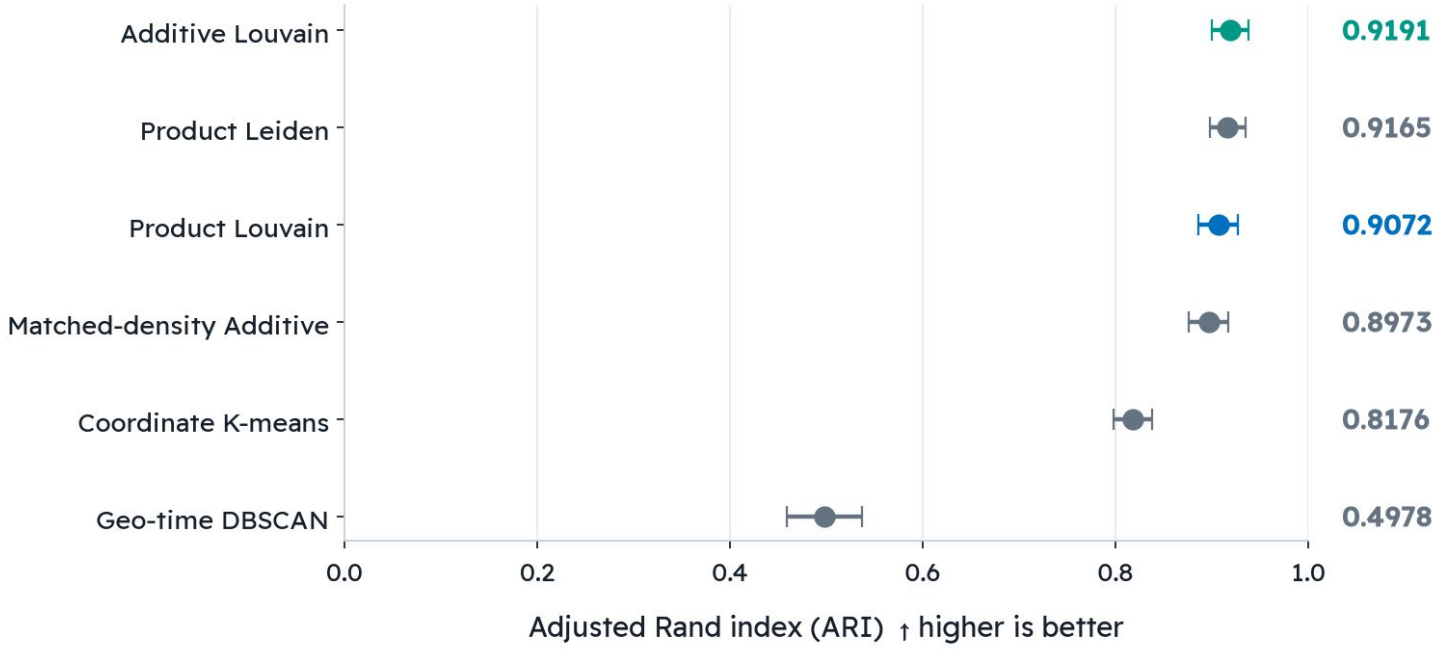
LOWER IS BETTER

Fraction of incidents whose simulated response time exceeds their own deadline.

Travel / service constraints → Locked schedule → Evaluator-only outcomes

RQ1: no Product composition advantage is established

Held-out Generator-3.0 test split · 40 paired runs



Additive Louvain

0.9191

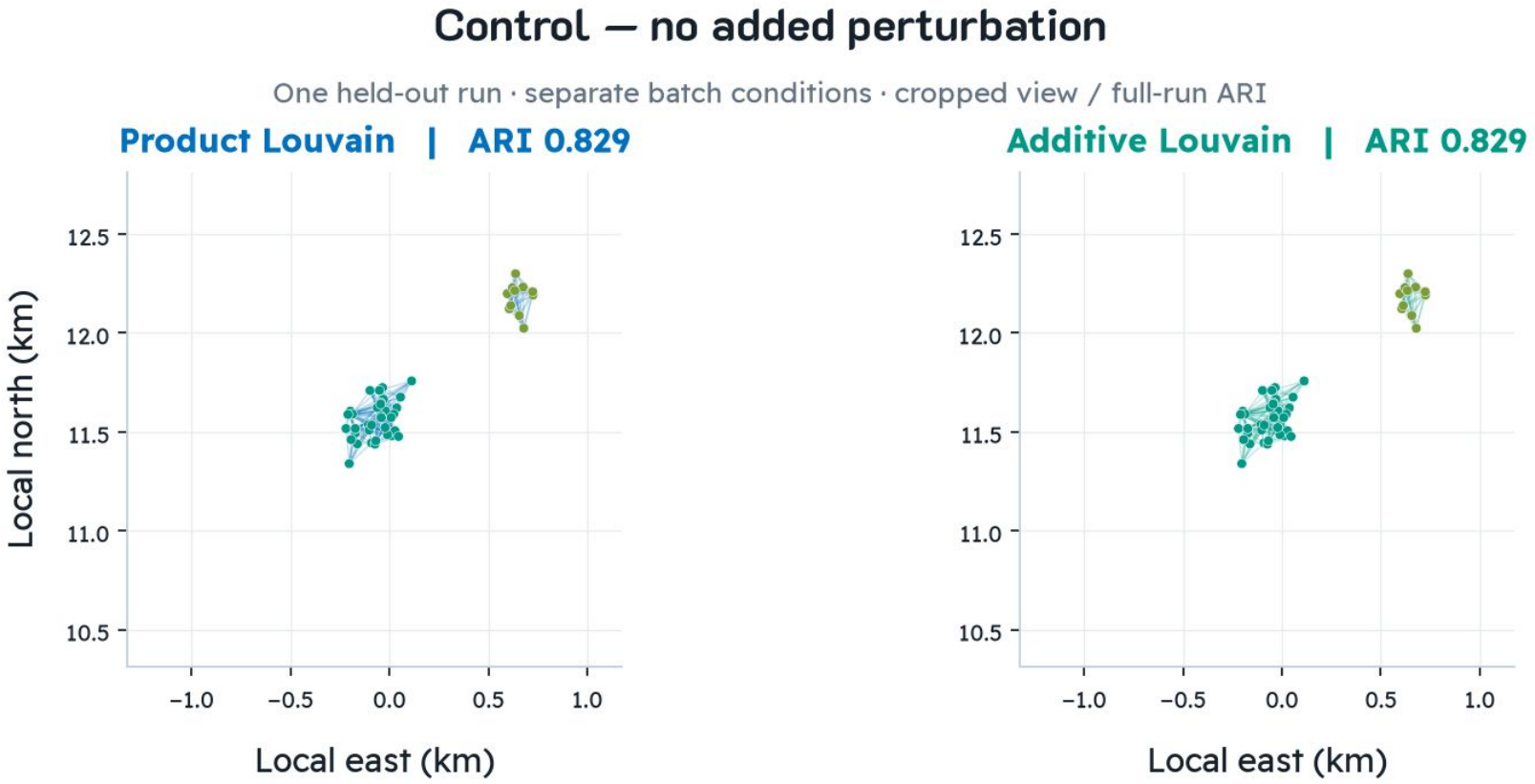
Product Louvain

0.9072

Holm-adjusted p = .2560 for the paired contrast

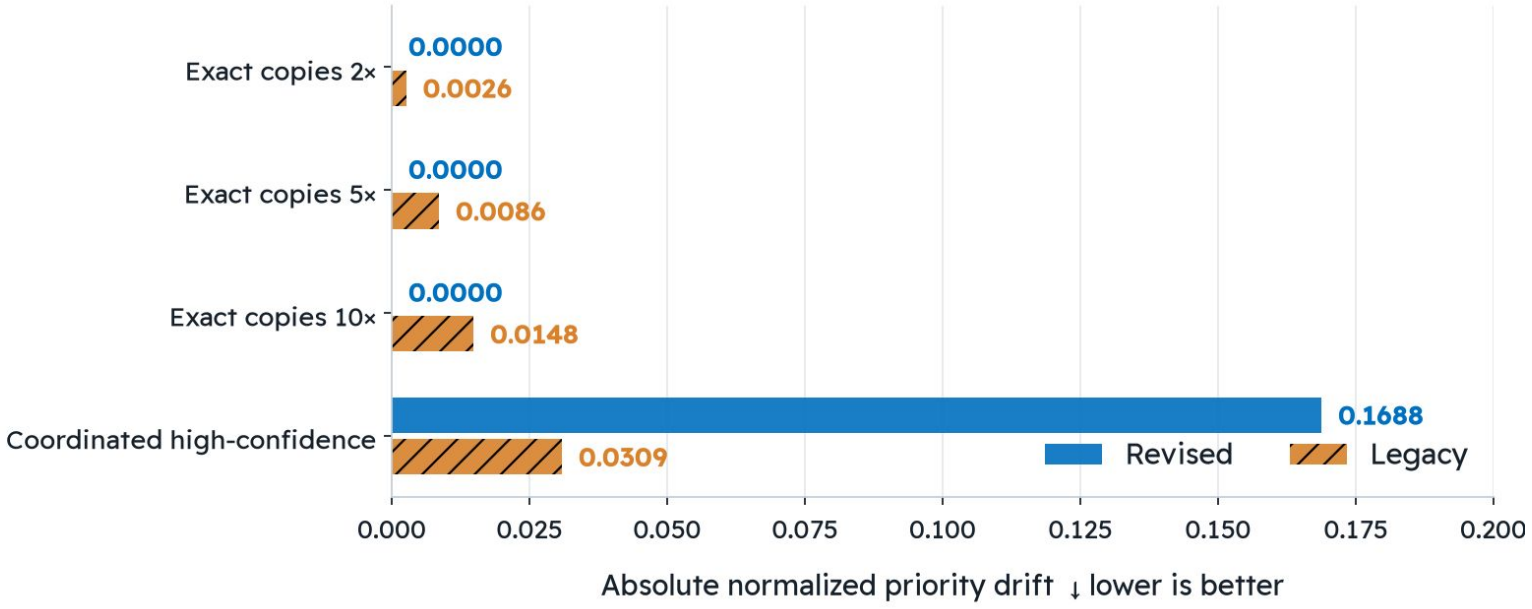
RQ1: graph structure changes under fixed stresses

Same run and locked configurations · separate perturbations of the control batch



RQ2: exact-copy stability has a narrow scope

Oracle incident groups and fixed score inputs



Exact copies

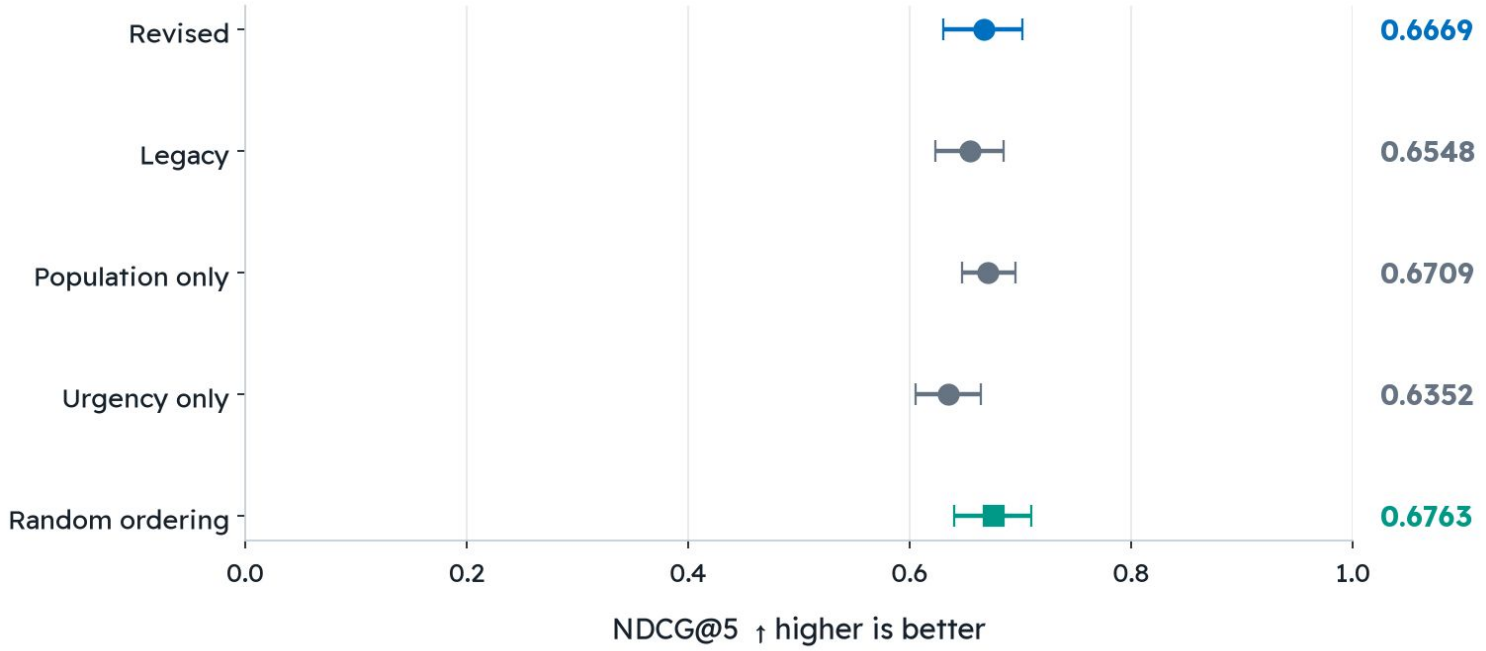
2× / 5× / 10×:
zero revised score drift.

Coordinated attack

High-confidence payloads
still distort priority.

RQ2: alignment with simulated benefit remains weak

Even oracle grouping does not establish a general ranking advantage



Revised NDCG@5

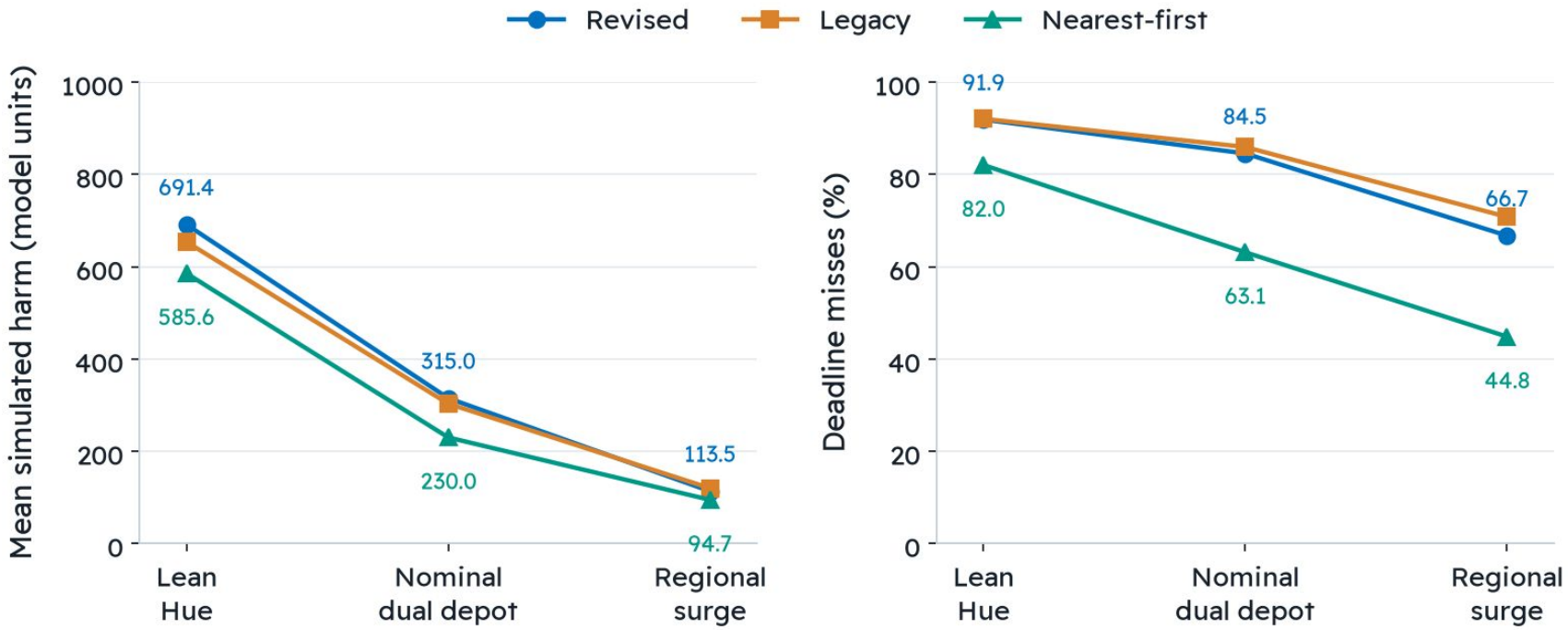
0.6669

Random comparator

0.6763

RQ3: safeguards do not ensure better dispatch

Product partition · three resource conditions · 40 test seeds



Discussion: three different levels of evidence

A guarantee at one stage does not validate the next stage

Mathematical safeguards

Conditional edge localization.

Exact-copy score invariance

under fixed grouping / inputs.

Observed synthetic results

Noise and 5× copies degrade

clustering; alignment is weak;

dispatch benefit is unestablished.

External validity

Two synthetic suites.

No expert-endorsed policy.

No field-outcome validation.

Conclusion

What this study establishes within its tested scope

1. **Product gating**

   Provides a conditional geographic edge bound; empirical composition superiority is not established.

2. **Family-wise priority**

   Is invariant to exact copies under fixed grouping and inputs; benefit alignment remains unestablished.

3. **Downstream audit**

   Reveals that clustering and ranking safeguards do not ensure improved simulated dispatch.

Future work: close the evidence gaps

Proposed validation steps, not results already obtained

01 Real report linkage

Acquire traceable report snapshots and independently annotate same-incident links.

02 Expert priority validation

Define and assess rescue priorities with domain experts and operational stakeholders.

03 Transfer and stress coverage

Freeze the pipeline and test new event contexts, missingness and source-quality mechanisms.

04 Scheduling objectives

Evaluate need-based priorities together with travel, service capacity and resource constraints.

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THANK YOU

Questions and discussion

Clustering safeguards, ranking stability and dispatch benefit require separate evaluation.

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Research artifact [12]: github.com/ngthtrong/nckh2/releases/tag/v1.0.1

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